MEERA KRISHNAN

Backend Engineer

INTERESTS

Long-distance running, classical guitar, competitive chess (state level, 2019). Volunteer instructor at a weekend coding club for school students since 2021. Member of the university photography society. Enjoys hiking and cooking.

LANGUAGES

English (fluent), Hindi (fluent), Tamil (native), German (basic).

PROFILE

Backend engineer who enjoys systems work and performance tuning.

EXPERIENCE

Backend Engineer, Northwind Systems
Jun 2022 - Present
Built and maintained the order-processing service handling roughly 4,000 requests per minute. Rewrote the pricing engine, cutting p99 latency from 840ms to 120ms by replacing repeated linear scans with an indexed lookup and an in-memory LRU cache with time-based eviction. Designed the Postgres schema for the settlements module, including partitioning by month and a set of covering indexes that removed three sequential scans from the nightly job. Wrote the migration tooling in Python. Owned the on-call rotation for the service.

Software Engineering Intern, Bluepeak Labs
Jan 2022 - May 2022
Implemented a shortest-path routing feature for the internal logistics tool using Dijkstra over a weighted graph of roughly 12,000 nodes. Reduced average route computation from 2.1s to 180ms by switching to a binary heap for the frontier. Wrote the service in Java with a Spring Boot HTTP layer.

PROJECTS

Distributed key-value store (personal, 2023)
Implemented a Raft-based replicated store in Java. Log compaction, leader election, snapshotting. Wrote a deterministic simulation harness to reproduce split-brain scenarios.

Text index for a 2GB corpus (personal, 2021)
Built an inverted index with a trie for prefix queries and a custom skip-list for postings intersection. Benchmarked against a naive scan.

EDUCATION

B.Tech Computer Science, Ashwood Institute of Technology, 2022
Relevant coursework: Design and Analysis of Algorithms, Database Management Systems, Operating Systems, Distributed Systems. CGPA 8.7/10.

TECHNICAL

Java, Python, Postgres, Redis, Docker, Kubernetes, Git, Linux, gRPC.